THOMAS MORE

Data Science Fundamentals

kMeans

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory exercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

(subject to change)

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Now we get to the fun part

kMeans

Clustering

- In clustering, only values of independent variables are given.
- We assume that there exists a certain dependent variable with different categories, **but we cannot observe this**.
- It is our task to find certain rules which help us to divide our dataset into different clusters.



Is clustering supervised or unsupervised?

Unsupervised!

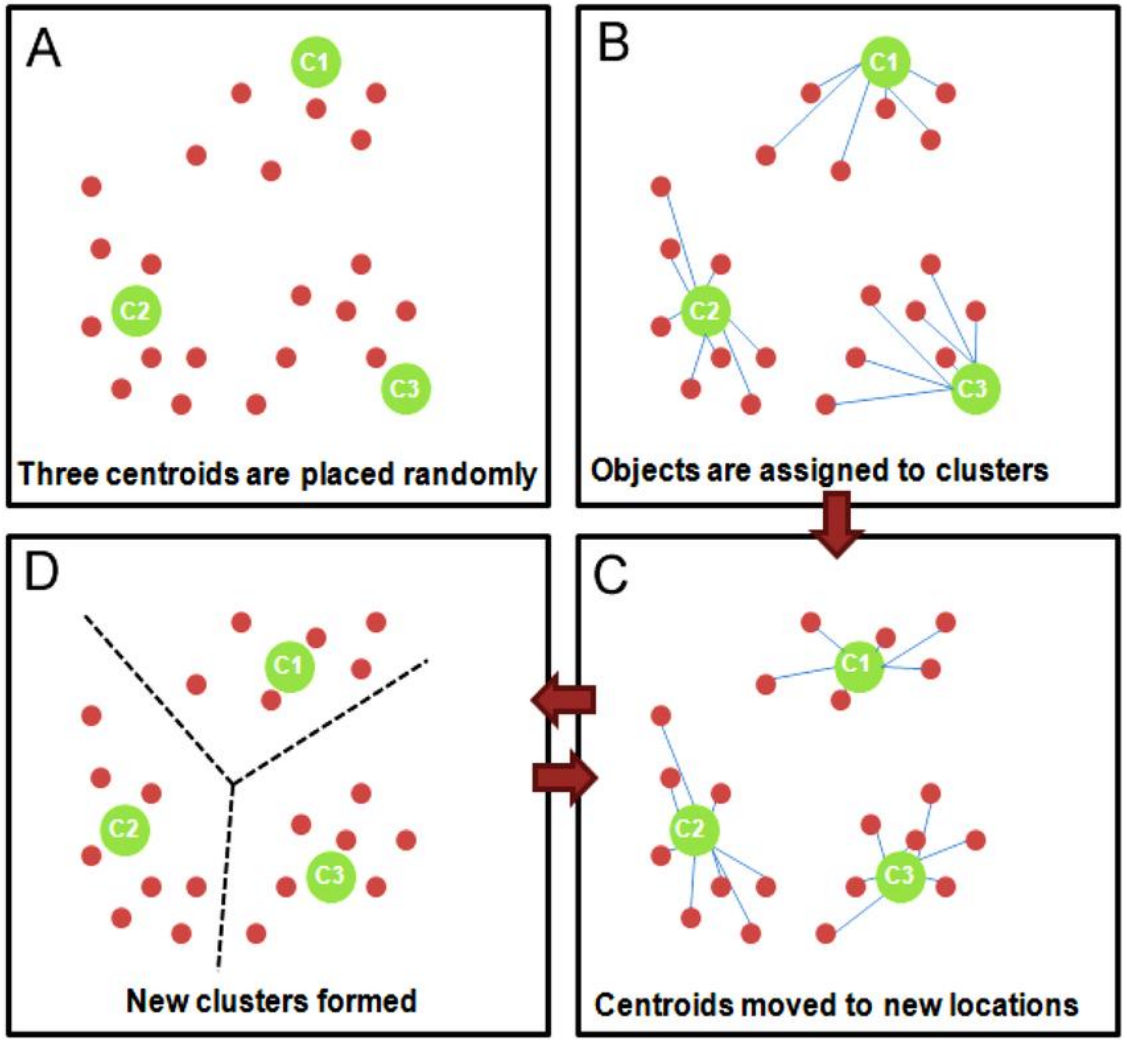
K means

- Basic clustering algorithm
- Start with a parameter ‘k’
- = amount of clusters
- Divide your dataset amongst each cluster
- Result: data is categorized without a specific label

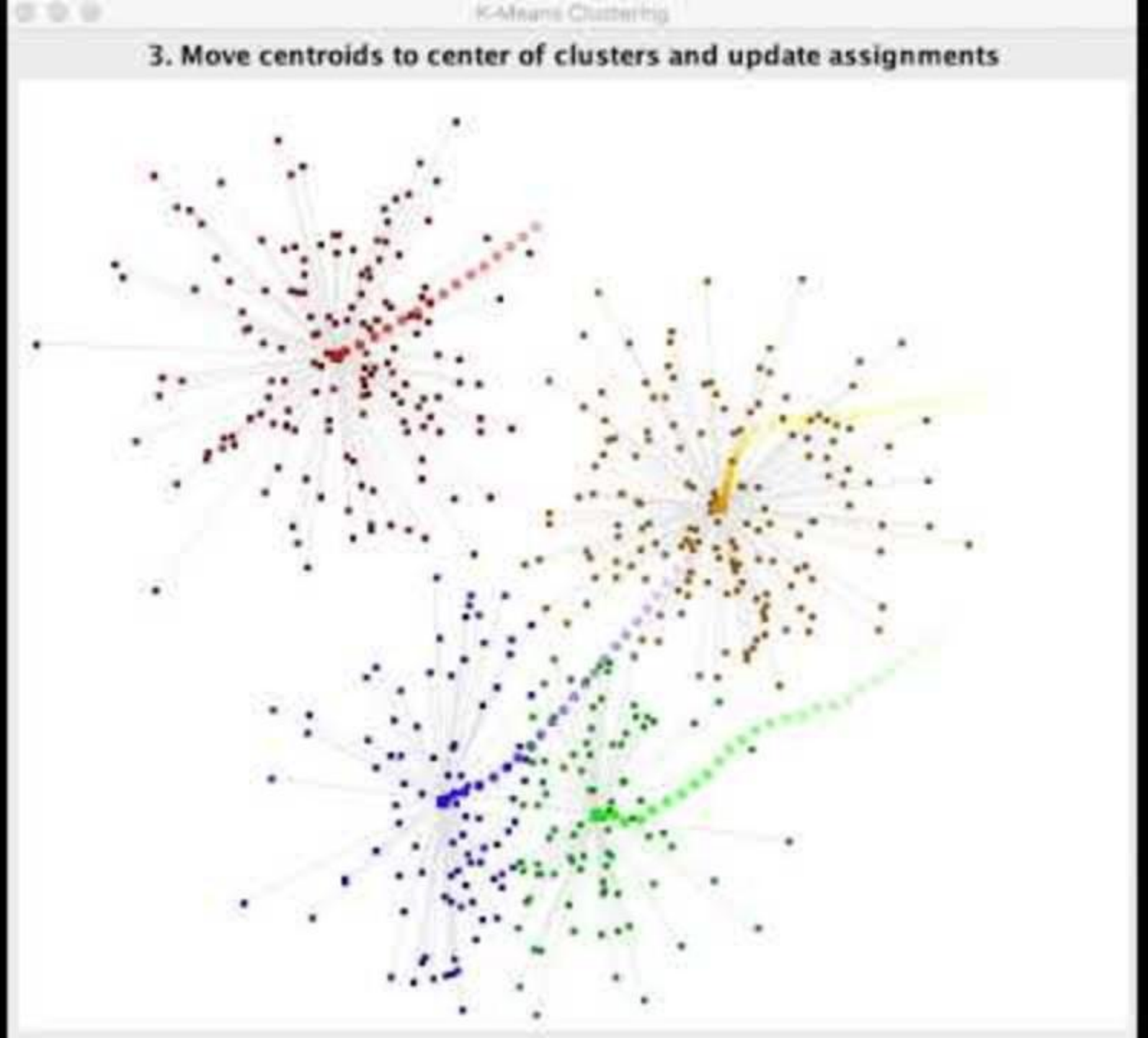


Inner workings

- Assign random starting point to each cluster (= ‘centroid’)
- Assign each datapoint to nearest centroid
- Move cluster to middle of each assigned datapoint
- Repeat until converged



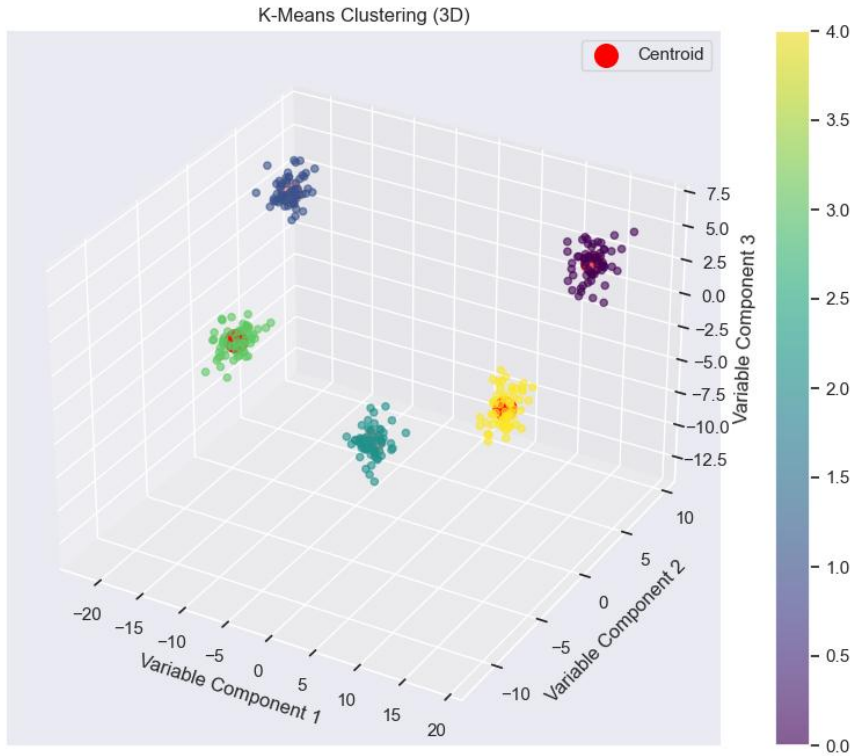
Source: Rumdeep K. Grewal, Microarray data analysis: Gaining biological insights (ResearchGate 2013)


K-Means Clustering
3. Move centroids to center of clusters and update assignments

Plotting the clusters

- In python, we can only plot this in 2D/3D.
- More than 3 columns/dimensions → dimensionality reduction & PCA
- We'll touch on that in a few lessons ;)

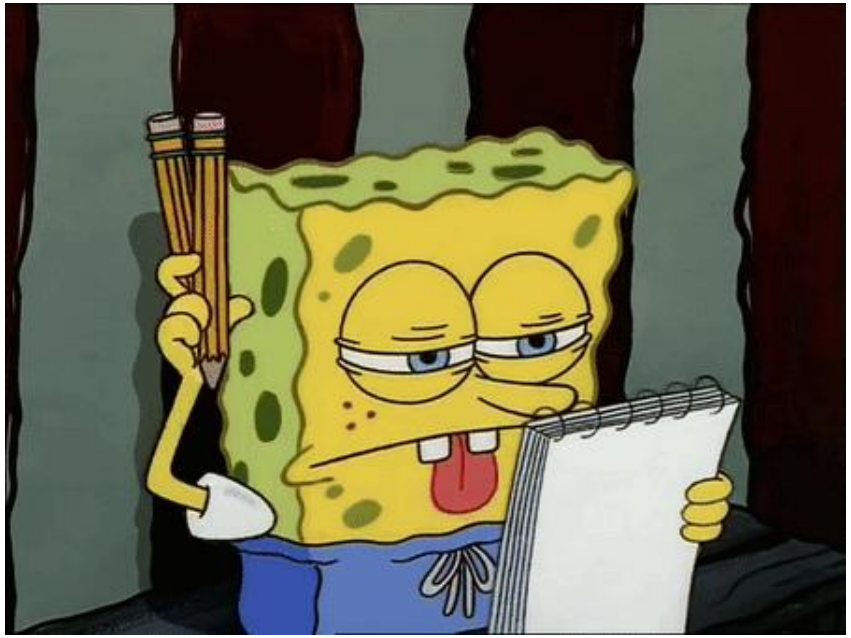




Finding the best k

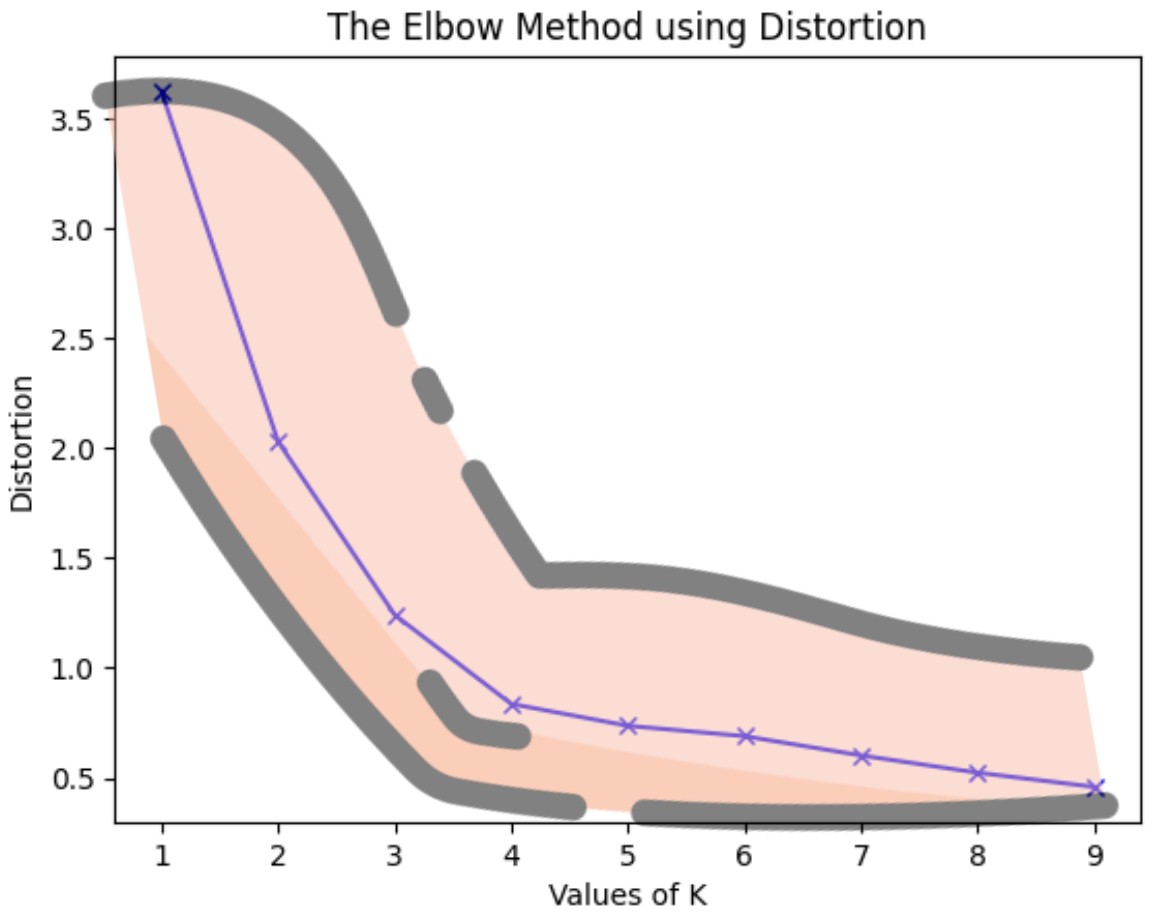
Sometimes we know what k to pick but most of the times we don’t.

Then how to we figure out which value of k to use?



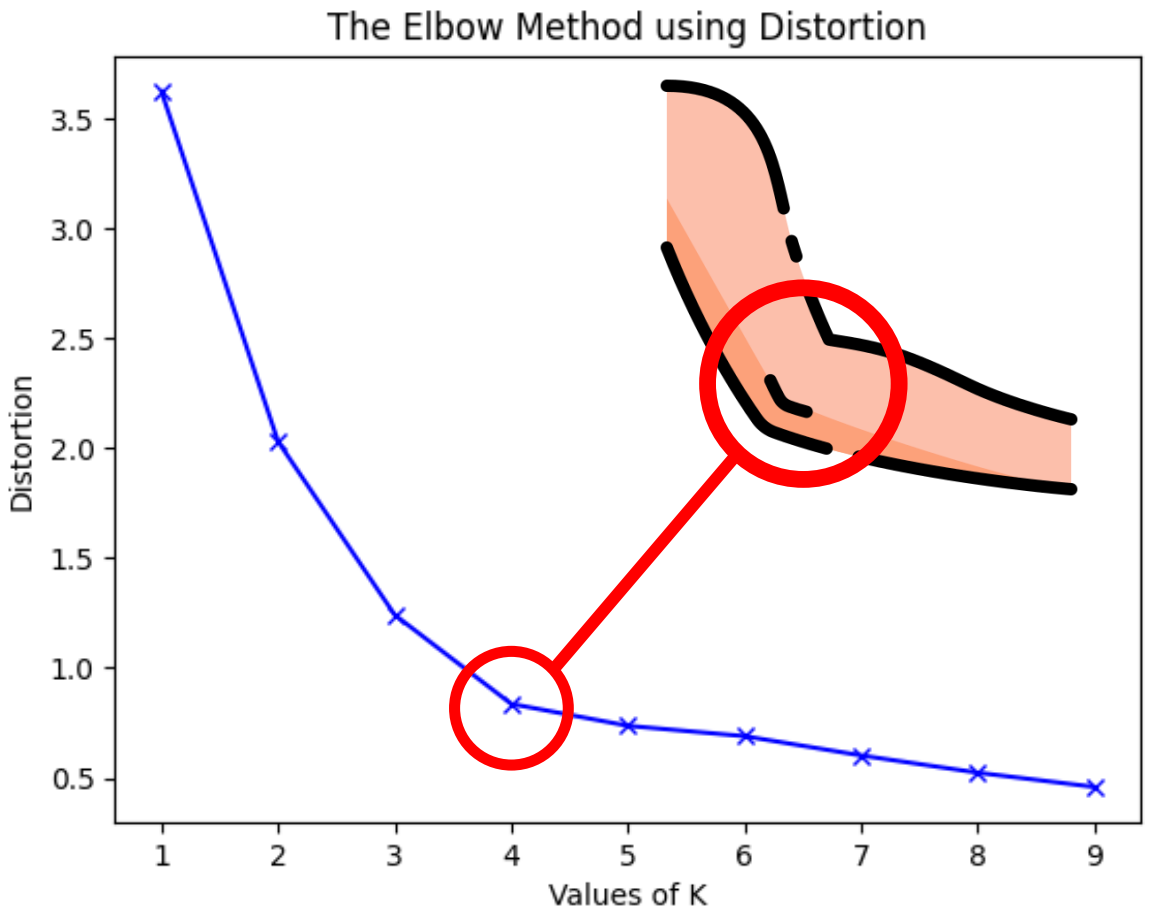
Elbow method

- Similar to finding the best value of k for knn
- Train a bunch of models
- Calculate '**distortion**' per model
- = average squared distance between each datapoint and its nearest centroid
- Find the right **balance** between low **distortion** and low value of **k**
- Pick the value in the middel of the 'elbow'



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Overfitting

Surely this time we don't have to worry about this since it's clustering right?

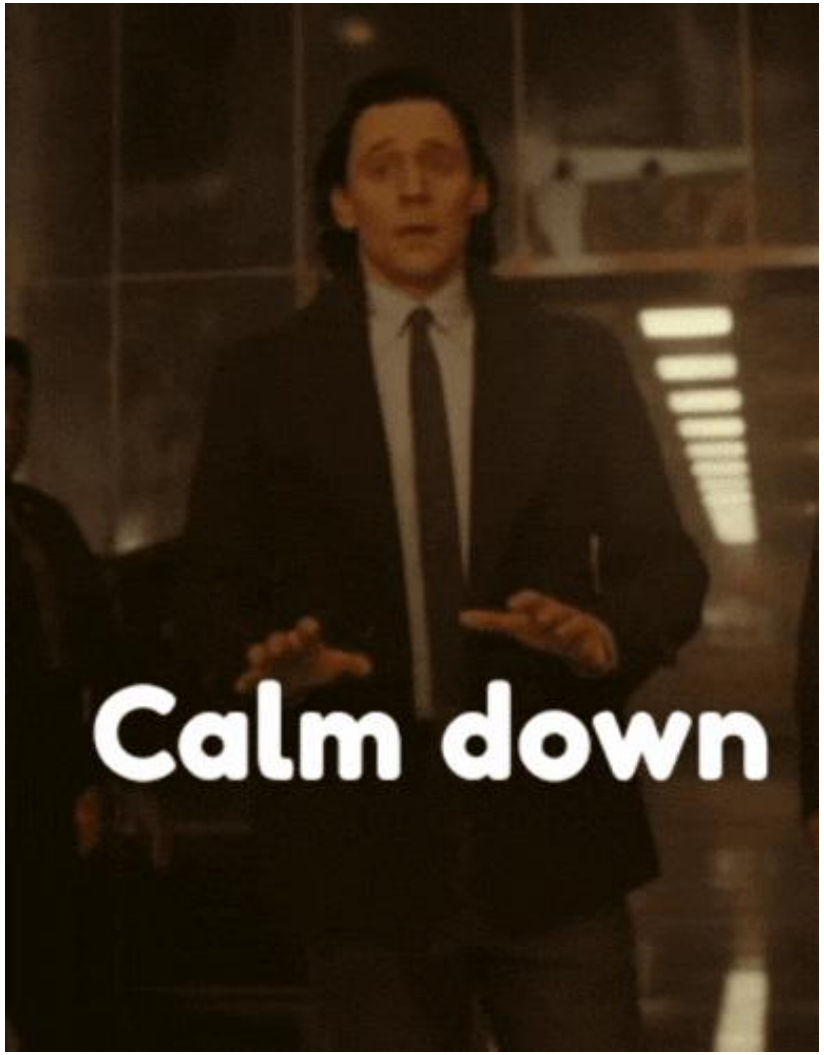
Technically, we do not *



*** Clustering can still overfit…**

Overfitting

- Clustering models can’t ‘overfit’ like regression or classification
- There is no label to memorize
- Overfitting occurs when you define **too many** clusters
- Underfitting occurs when you define **too few** clusters
- Just use the elbow method and you’ll be fine ;)



Evaluation metrics

- Since we have no labels, there is no way to evaluate our model
- The only way to confirm is by plotting your data
- If you limit you data to two columns, you can create a graph

- Secret technique: we can ‘mold’ our clustering model to produce results



Clustering as a predictive model

- Strip your dataset of labels
- Perform clustering
- Assign each datapoint to a cluster
- Map each cluster to a label
  - Look at which class each cluster is **most likely** to represent
  - =/= 100% reliable!
- Compare ‘predicted’ label with stripped label
  - Accuracy, precision, correlation matrix, …
  - Clustering → no need for a train test split!
- Done!



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Let's jump into action

Excercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 07 – kMeans Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 07 – kMeans Exercise"
- Also download the csv files called "Mall_Customers.csv" and "Wine.csv" (from chapter 5)
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises



I don't bite, ask me any questions you have!